

Hibernica - Permissionless Intent-Based Options

Infrastructure Layer for Tokenized Assets, built by the Ivy Foundation.

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Abstract

Hibernica is a permissionless options infrastructure layer designed to transform any tokenized asset into an investable, yield-generating, and risk-transferable options market. Where spot markets became permissionless through automated market makers, the equivalent transition has not yet occurred for derivatives. Hibernica addresses this structural gap by enabling holders, whales, communities, foundations, and eventually any tokenized asset issuer to create option markets without requiring bespoke bilateral negotiations, closed market-maker relationships, or centralized listing processes.

The product begins with a simple proposition: any holder of an asset should be able to monetize its volatility, and any participant with a pricing model should be able to bid for that volatility. Hibernica turns idle inventory into structured yield opportunities, while giving sophisticated counterparties access to statistical edge across a broad universe of assets.

Complementing Hibernica is Helix, Hibernica's market-maker white-label solution. Helix packages the same underlying infrastructure into a professional-grade quoting and execution stack for market makers, allowing them to stream, manage, and monetize options liquidity across large asset universes, strikes, expiries, and market configurations.

Together, Hibernica and Helix create a unified architecture for the next phase of derivatives: permissionless origination, competitive pricing, programmable execution, spider-web quoting across an infinite amount of potential strikes and institutional-grade market-making at internet scale.

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1. Executive Summary

Hibernica is a permissionless options-market infrastructure layer for tokenized assets. It enables any holder, foundation, whale, or community to create an options market by depositing inventory into a vault, while allowing market makers and sophisticated counterparties to competitively bid for that volatility. The system transforms idle token inventory into yield-generating option supply and gives liquidity providers scalable access to long-tail volatility opportunities.

Its core innovation is market origination. Rather than relying on centralized listing committees or bilateral OTC negotiations, Hibernica allows markets to emerge wherever inventory and pricing demand exist. This makes it particularly suited for the long tail of crypto assets and the future universe of tokenized real-world assets.

Helix is Hibernica’s white-label market-maker solution. It packages the same infrastructure into a professional-grade quoting and execution stack that allows market makers to operate branded options platforms, stream prices across large asset universes, enforce model-driven execution constraints, and scale their volatility business without building the entire venue infrastructure themselves.

Together, Hibernica and Helix form a complementary architecture: Hibernica permissionlessly creates the markets; Helix institutionalizes the liquidity layer around them. The result is a scalable framework for options on any tokenized asset.

2. The Problem: Tokenization Without Derivatives Is Incomplete

Over the last decade, crypto markets have demonstrated the viability of permissionless spot-market formation. Automated market makers, most notably Uniswap, showed that new assets can become tradable when market creation is open, composable, and operationally simple. This development substantially reduced the dependency on centralized listing processes and bilateral liquidity arrangements for spot assets.

Derivatives markets, however, have not undergone an equivalent transition. While BTC, ETH, and a small number of highly liquid crypto assets have developed active options markets, the vast majority of tokens issued through decentralized venues remain without functional, smart-contract-native options infrastructure. Where options exposure exists for long-tail assets, it is typically arranged through bespoke bilateral agreements, centralized venues, or legally enforced over-the-counter contracts rather than through standardized, programmable, and permissionless smart-contract systems.

This creates a structural asymmetry in on-chain finance. Token issuance and spot-market formation are increasingly permissionless, but volatility markets, hedging instruments, yield strategies, and convex payoff structures remain accessible only for a narrow subset of assets. As a result, most tokenized assets can be transferred and traded, but they cannot be efficiently hedged, optioned, or transformed into structured yield products.

The current options market therefore remains constrained by several inefficiencies.

2.1. Closed Access

Options markets are usually created through private relationships between foundations, funds, whales, and market makers. Access is negotiated, opaque, and limited to a small set of large participants.

2.2. Illiquidity in the Long Tail

Thousands of tokens have active communities, meaningful holder bases, and volatility, but no functional options markets. These assets may have spot liquidity, and sometimes perpetual futures liquidity, but no efficient way for holders to sell volatility or for traders to express convex views.

2.3. Inefficient OTC Workflows

Today's OTC option market is relationship-driven and operationally heavy. A foundation or whale negotiates with one or two counterparties, receives a bespoke quote, and executes bilaterally.

2.4. Idle Inventory

Token holders, foundations, whales, and communities often sit on large inventories that produce no yield. Their only practical choices are to hold, sell, lend, or engage in highly punitive OTC structures.

2.5. No Scalable Infrastructure for Tokenized Assets

As more assets become tokenized, the natural question becomes: what happens next? Once an asset is tokenized, it requires markets, risk transfer, liquidity, leverage, and options. Without derivatives, tokenization remains an limited issuance technology rather than a full financial system.

Hibernica is built around the thesis that every tokenized asset will eventually need an options market, and that this market cannot be created through manual, bilateral, institution-by-institution workflows.

3. Hibernica: Permissionless Options for Any Asset

Hibernica is a permissionless options platform where any eligible asset holder can deposit tokens into a vault and expose that inventory to competitive option bidding. Market makers, sophisticated traders, and other counterparties can submit bids for options on that inventory. The system then facilitates execution according to the depositor's constraints, target yield, and market conditions.

The core primitive is simple:

Depositors supply inventory.
Bidders supply option premiums.
Hibernica coordinates execution.

This creates a market where long-tail assets can become optionable without waiting for centralized venues, large exchanges, or exclusive market-maker arrangements.

3.1. Core Proposition

Hibernica allows any token holder to transform idle inventory into an options vault. This inventory can then be monetized through covered-call-style structures or other option configurations. The depositor remains passive: they define broad economic preferences, such as target APR, eligible duration, or acceptable moneyiness, while the system sources and evaluates bids.

The bidder receives access to a structured option position with clearly defined terms. The depositor receives premium income. Hibernica coordinates the market.

In its first form, Hibernica is focused on call-selling inventory: whales, foundations, or communities deposit tokens, while market makers bid for the right to buy options against that inventory.

4. Hibernica - open market creation

Hibernica introduces to options what automated market makers introduced to spot markets: open market creation.

Today, if a token does not have an options market, the community has few realistic ways to create one. A foundation may need to convince a centralized venue, negotiate with a market maker, or wait until the asset reaches sufficient size. Hibernica changes this dynamic.

A market can be created by:

- the foundation,
- a whale,
- a community member,
- a market maker,
- a tokenized asset issuer,
- or any actor holding inventory.

This is a meaningful structural shift. It moves option-market origination away from closed institutional coordination and toward open, permissionless market formation.

The result is a system where the long tail of tokenized assets can become financially productive.

5. The Hibernica Market Structure

Hibernica can be understood as a three-sided marketplace.

5.1. Inventory Providers

Inventory providers are actors who deposit assets into Hibernica vaults. Their goal is simple: earn premium income without manually negotiating bespoke option deals.

Instead of setting complex option parameters themselves, depositors can express their preferences through intuitive inputs such as target APR, duration, and risk tolerance. Hibernica can then translate those preferences into executable option logic.

5.2. Bidders and Market Makers

Bidders are sophisticated counterparties seeking edge.

Market makers do not necessarily need to source assets, negotiate with foundations, or maintain exclusive OTC relationships. Hibernica gives them a broad surface area of potential option opportunities. They can inspect vaults, price risk using their own models, and submit bids.

The attraction for market makers is not merely one market. It is the ability to access an expanding universe of assets, strikes, expiries, and volatility opportunities.

In institutional language, Hibernica creates a scalable inventory-discovery layer for volatility buyers, speeding up deal making trade matching.

5.3. The Solver and Execution Layer

The solver layer coordinates matching and execution. In early phases, this may be more curated or semi-centralized, allowing Hibernica to ensure execution quality, avoid economically irrational fills, and learn from live market behavior.

Over time, this execution layer can become more automated, more rule-based, and potentially more decentralized. The guiding principle is best execution: if a depositor is willing to sell volatility at a minimum economic threshold, the system should seek to execute above that threshold and return the surplus benefit to the depositor rather than allowing an intermediary to extract it opaquely.

This best-execution solver concept is critical. It allows Hibernica to position itself not as a broker extracting hidden spread, but as an execution coordinator designed to improve outcomes for

passive inventory providers.

6. A Simple Example

A whale holds a large amount of *Token X*. They do not want to sell the asset outright, but they are willing to sell upside above a certain level in exchange for premium income.

Through Hibernica, they deposit *Token X* into a vault and define a target yield. Hibernica presents the vault to potential bidders. Market makers submit option bids. Hibernica evaluates the bids against the depositor's target and selects the best executable terms.

The whale earns premium.

The market maker receives an option with edge.

The token ecosystem gains a functional options market.

The asset becomes more financially useful.

Trading begets trading.

This is the core Hibernica primitive.

7. Why Permissionless Options Are Strategically Important

The long-term opportunity is not limited to existing crypto tokens.

The broader financial market is moving toward tokenization. Equities, bonds, funds, private credit, real estate yield, structured products, and other claims are increasingly being represented on-chain. But tokenizing an asset is only step one. A tokenized asset without derivatives is financially incomplete.

Once an asset is tokenized, market participants will want to:

- buy it,
- sell it,
- short it,
- hedge it,
- borrow against it,
- lend it,
- price its volatility,
- and create options on it.

Hibernica positions itself as the options layer for this future.

If an asset can be tokenized, Hibernica aims to make it optionable.

The central academic thesis is:

Tokenization standardizes the representation of value. Hibernica standardizes the creation of volatility markets around that value.

8. Differentiation Against Existing Options Venues

Hibernica is not simply another options venue. Its differentiation lies in market origination. Existing venues may offer options on selected assets, usually large-cap assets or assets with existing institutional demand. Hibernica is designed for the opposite direction: market creation from the inventory side.

Rather than asking, “Which assets should an exchange list?” Hibernica asks:

Who has inventory, and who wants to bid for its volatility?

This inversion matters. It enables the creation of markets before traditional venues would consider them viable.

Hibernica’s key differentiators are:

1. **Permissionless market creation.** Any holder can initiate an options market by depositing inventory.
2. **Long-tail asset coverage.** Hibernica is designed for assets that do not yet have mature options markets.
3. **Private economic preferences.** Depositors can define target APR or minimum acceptable economics without exposing all sensitive information to bidders.
4. **Competitive bidding.** Market makers compete for inventory rather than receiving exclusive bilateral access.
5. **Best-execution orientation.** Hibernica can route execution toward the best available bid relative to depositor constraints.
6. **Scalable quote surface.** The architecture can support very large numbers of assets, strikes, and expiries.
7. **Compatibility with tokenized assets.** Hibernica is not limited to crypto-native tokens. The same logic can extend to any asset represented on-chain.

9. The Yield Primitive

For communities and whales, Hibernica creates a new yield primitive.

Most token holders are structurally long. They hold assets through volatility, often without access to sophisticated monetization tools. Covered-call programs are common in traditional finance, but in crypto they remain difficult to access outside curated vault products or OTC relationships.

Hibernica democratizes this structure.

A community no longer needs to wait for a centralized party to provide yield. A whale or foundation can open a market, deposit inventory, and invite market makers to bid. If the asset has volatility and counterparties are willing to pay for upside exposure, a yield market can emerge.

This is especially powerful for communities whose holders are underwater, inactive, or searching for productive use cases for their tokens. Hibernica gives those communities an answer:

Your asset can become productive without being sold.

10. Market-Maker Incentive

For market makers, Hibernica offers access to edge.

The market-maker business is fundamentally about pricing risk, finding mispriced volatility, and deploying models across many opportunities. Hibernica increases the surface area of possible opportunities dramatically.

Instead of relying on private sourcing, relationship-based foundation deals, or centralized venue listings, a market maker can access a broad inventory network where holders voluntarily expose assets for option bidding.

The more markets Hibernica lists, the more valuable the platform becomes to sophisticated counterparties.

For a market maker, Hibernica can be framed as:

A long-tail volatility discovery engine.

It surfaces assets, vaults, strikes, expiries, and inventory pockets that would otherwise remain inaccessible or operationally expensive to source.

11. Helix: Hibernica's Market-Maker White-Label Solution

While Hibernica is the public permissionless options marketplace, Helix is the professional white-label infrastructure layer for market makers.

Helix packages the underlying Hibernica technology into an institutional product that allows market makers to operate their own branded options environment, quote across large market surfaces, and integrate their proprietary pricing, risk, and execution workflows.

Where Hibernica opens the market to everyone, Helix gives professional liquidity providers the machinery to industrialize their participation.

11.1. Helix Value Proposition

Helix allows a market maker to deploy a customized options platform without building the full infrastructure stack from scratch.

A market maker using Helix can:

- quote across many assets,
- support many strikes and expiries,
- integrate proprietary pricing models,
- manage custom execution constraints,
- expose selected markets to clients,
- support vault-based inventory sourcing,
- and operate under its own brand or institutional workflow.

In essence:

Helix turns Hibernica’s permissionless options infrastructure into a white-label volatility platform for professional market makers, to quote full surfaces on N assets as easy as providing a Limit on an Orderbook.

11.2. Why Market Makers Need Helix

Market makers are facing a strategic transition. The number of tokenized assets is growing faster than centralized venues can list them, and faster than bespoke OTC desks can cover them manually.

At the same time, competition for obvious markets is increasing. Large-cap assets are crowded. Margins compress. Edge moves to less efficient markets, niche assets, and fragmented inventory.

Helix gives market makers a way to scale into this opportunity.

Instead of building venue infrastructure, vault management, intent handling, execution logic, quote interfaces, and settlement flows internally, they can use Helix as a white-label stack.

This lets them focus on what they do best:

Pricing risk and monetizing edge.

11.3. Helix as a Quote-Surface Engine

The central technical strength of Helix is its ability to support a massive quote surface.

A traditional venue may list a limited number of assets, strikes, and expiries. Helix is designed for the opposite problem: an environment where millions of assets can exist, where any tokenized claim may require an options market, and where users may demand custom strikes or expiries.

Helix is therefore not merely a trading interface. It is a quote-surface engine.

It allows professional market makers to operate across an asset universe that is too large for traditional listing committees, manual OTC desks, or centralized product teams.

This is particularly important in a world where tokenization produces extreme asset proliferation.

11.4. Model Privacy and Execution Control

A critical design requirement for Helix is that market makers must be able to protect their models.

Market makers do not want to reveal proprietary pricing logic. They want to submit bids, enforce conditions, and protect themselves against stale quotes, price moves, volatility shifts, or other risk events.

Helix can support this through conditional intents, verifiers, and oracle-like validation logic. A market maker can submit quotes subject to specific execution constraints. These constraints may relate to price bands, volatility assumptions, time windows, risk parameters, or other model-derived values.

The key idea is that the market maker does not need to reveal the model itself. Instead, the system can verify whether execution conditions are satisfied.

This creates a flexible architecture:

The model remains private.
The execution constraints become verifiable.
The market opens to new actors.

12. Hibernica and Helix Together

Hibernica and Helix are the same product offered differently.

Hibernica creates public market formation.

Helix gives professional market makers the institutional tooling to participate at scale.

Product	Primary User	Core Function
Hibernica	Whales, foundations, communities, tokenized asset issuers	Create permissionless options markets and monetize inventory
Helix	Market makers, trading firms, institutional liquidity providers	White-label quoting, execution, and options-market infrastructure

Hibernica creates the inventory network.

Helix monetizes the liquidity provider network.

Hibernica answers the question:

How does any asset get an options market?

Helix answers the question:

How does a market maker quote and scale across all of them?

13. Strategic Narrative

The broader narrative is simple.

First, crypto made assets transferable.

Then automated market makers made spot markets permissionless.

Then perpetuals made leverage broadly accessible.

The missing layer is permissionless options.

Hibernica is designed to become that missing layer.

As tokenization expands, the number of assets requiring financial markets will grow dramatically. Manual listing processes cannot scale to that world. Bilateral OTC relationships cannot scale to that world. Centralized venues cannot list every meaningful market.

A permissionless options protocol can.

Helix then extends this into the institutional domain by giving market makers the tools to commercialize this opportunity directly.

Together, Hibernica and Helix create a two-sided strategy:

1. Capture inventory from the long tail of token holders and tokenized asset issuers.
2. Enable professional liquidity providers to price and monetize that inventory at scale.

This forms a self-reinforcing loop:

More assets create more vaults.
More vaults attract more bidders.
More bidders improve execution.
Better execution attracts more depositors.
More market activity increases the value of the infrastructure.

14. Technical Overview: How Hibernica Works

Hibernica is technically a vault-based OTC options marketplace with permissionless market creation, private bid submission, and a solver-controlled execution layer.

At the technical level, it has four core components:

1. Vaults and inventory deposits.
2. Market-maker bids and signed intents.
3. A solver or matching layer.
4. Option and settlement execution on Hibernica infrastructure.

The simplest technical definition is:

Hibernica is a permissionless options-origination system where token holders deposit inventory into vaults, market makers submit signed conditional bids to buy options on that inventory, and a solver executes the best valid bids against private depositor constraints, distributing option premium back to depositors as yield.

14.1. Step One: Asset Holders Deposit Inventory Into a Vault

The starting point is the seller side.

A token holder, whale, foundation, community member, or market creator deposits tokens into a Hibernica vault.

For example, a whale holds 1,000,000 units of Token X. They deposit a portion of those tokens into a Hibernica vault. That vault now represents optionable inventory.

The depositor does not need to manually structure a full options book. They should not be required to say:

“I want to sell a 30-day 25-delta call at strike X for premium Y.”

Instead, the system should abstract this into simpler parameters.

From a product perspective, the preferred input is a target APR. The depositor effectively says:

“I am willing to sell upside on this inventory if the resulting yield is at least X percent APR.”

Potential additional parameters may include:

- asset deposited,
- amount deposited,
- duration preference,
- rough moneyness or out-of-the-money preference,
- maximum utilization,
- accepted collateral or settlement asset,
- expiry window,
- withdrawal constraints,
- and risk tolerance.

The depositor is passive. They deposit inventory and define economic constraints.

14.2. Step Two: The Depositor’s Target Economics May Be Hidden

A key design idea is that the depositor’s minimum acceptable economics should not be public.

If the market maker knows the depositor’s minimum acceptable APR, the market maker can bid just above that threshold and capture the entire surplus.

Therefore, the intended flow is:

1. The depositor enters a target APR.
2. That APR is encrypted, committed, or otherwise hidden.
3. The value is stored or referenced in a way that only the Hibernica solver can use.
4. Market makers cannot see the threshold.

This gives the vault a private reservation price. Economically, it is the seller’s minimum acceptable premium.

Technically, this could be implemented through:

- encrypted calldata,
- off-chain storage with on-chain commitments,
- threshold encryption,
- solver-held decryption keys,
- trusted execution environments,
- or eventually a more decentralized private execution environment.

For the first version, the simplest acceptable version may be that the solver knows the private target APR and uses it during matching.

14.3. Step Three: Market Makers Submit Bids as Signed Intent

On the other side, market makers submit bids.

A bid is not merely an indicative quote. It should be a binding signed intent.

The market maker says:

“I am willing to buy a call option on this asset under these parameters, and I will pay this premium.”

The bid likely includes:

- asset,
- vault or market identifier,
- option type, initially likely call options,
- notional size,
- expiry,
- strike or strike rule,
- premium,
- collateral asset,
- validity period,
- price-band constraints,
- volatility, delta, or moneyness constraints,
- oracle or reference price requirements,

- and a signature from the market maker wallet.

The market maker must also have the premium available in the wallet or escrow flow.

A simplified conceptual structure is:

```
struct OptionBidIntent {
    address bidder;
    address vault;
    address underlying;
    address premiumAsset;
    uint256 notional;
    uint256 strike;
    uint256 expiry;
    uint256 premium;
    uint256 validUntil;
    bytes executionConstraints;
    bytes signature;
}
```

The exact production schema may differ, but conceptually the bid is a signed executable order with attached execution constraints.

14.4. Step Four: Bids Include Execution Constraints

Market makers need quote protection.

If a market maker submits a bid and then the underlying price moves significantly, they may no longer want that quote executed. Therefore, bids should not only contain a premium. They should contain conditions.

Examples include:

- only valid if spot price is between X and Y,
- only valid until timestamp T,
- only valid if implied volatility is within a range,
- only valid if oracle price is fresh,
- only valid if liquidity, spread, or index conditions are satisfied,
- only valid for a maximum notional,
- only valid if the option remains within a certain moneyness range.
- we are providing a turing complete oracle infra, where anyone can develop their own conditions making a theoretical infinite amount of conditions tradeable.

This is where Hibernica becomes a conditional intent system rather than a simple order book.

The bid says:

“Execute this option trade only if my conditions are satisfied at execution time.”

Technically, those conditions can be verified by:

- an on-chain oracle,
- an off-chain solver,
- a market-maker-provided verifier,
- a registered oracle contract,
- or a later trusted execution or verification system.

One proposed design is that the market maker can register an oracle or verifier that returns true or false for whether the quote is still valid. This protects their proprietary model. They do not expose the full model; they only expose an execution-validity check.

The market maker does not reveal:

“Here is my volatility surface and pricing model.”

Instead, they provide:

“Here is a verifier that confirms whether this quote is executable under my current model constraints.”

14.5. Step Five: Hibernica Collects Bids Privately

Market makers submit bids into the system. These bids may be displayed, partially displayed, or kept private depending on the product design.

The preferred direction is private competitive bidding, at least for sensitive values.

The reason is to prevent bid-sniping, threshold leakage, and unnecessary information disclosure.

The system may therefore have:

- public vault discovery,
- private bid submission,
- private depositor thresholds,

- solver-side bid comparison,
- and selective execution.

The public can see that a vault exists. For example:

“Token X vault: 1,000,000 Token X available for covered-call strategies.”

But market makers may not see:

- the depositor’s minimum APR,
- other market makers’ exact bids,
- full execution thresholds,
- or the solver’s ranking logic.

This preserves competitive dynamics.

14.6. Step Six: The Solver Evaluates Bids Against Vault Constraints

This is the key matching step.

The solver has access to:

- vault inventory,
- depositor constraints,
- hidden target APR,
- submitted market-maker bids,
- current oracle prices,
- expiry and moneyness information,
- risk checks,
- available notional,
- and system rules.

The solver asks:

Which bids satisfy the depositor’s minimum conditions and provide the best execution?

For example:

Bidder	APR Equivalent
Depositor Target	30%
Market Maker A	35%
Market Maker B	42%
Market Maker C	28%

The solver rejects Market Maker C and accepts Market Maker B first. If there is remaining capacity, it may then accept Market Maker A.

This is the BestEx or best-execution solver.

The principle is:

Do not simply fill at the depositor's minimum. Execute at the best available market-maker bid and return the surplus to the depositor.

This is important because otherwise the system would behave like a broker capturing hidden spread. Hibernica should instead behave as an execution system that improves depositor outcomes.

14.7. Step Seven: Accept Bid, Create Option Positions

Once the solver accepts a bid, the system creates the actual option exposure.

In the simplest first-version covered-call model:

- the vault has deposited tokens,
- the market maker pays premium,
- the market maker receives a contractual synthetic call option,
- the vault is obligated to deliver tokens or settle value if the option expires in the money.

There are two broad settlement models.

14.8. Physical Settlement

If the call finishes in the money, the option buyer can buy the underlying tokens at the strike price. The vault's inventory is used to deliver the token.

14.9. Cash Settlement

The system computes the payoff:

$$\max(S_T - K, 0) \times Q$$

where:

- S_T is the underlying price at expiry,
- K is the strike price,
- Q is the option quantity.

The payoff is then settled in a collateral asset.

For long-tail tokens, physical settlement may be attractive because the vault already holds the token. For more complex or tokenized assets, cash settlement may be easier.

14.10. Step Eight: Premium Flows to the Vault

The market maker pays a premium.

That premium is distributed to vault participants, minus protocol, solver, and market-creator fees.

Conceptually:

Market maker premium

```
-> protocol fee / solver fee / market creator fee
-> remaining premium to vault depositors
```

Depositors experience this as yield.

The user interface can express this yield as APR.

14.11. Step Nine: MVP Can Be semi-Manual, semi-centralized execution of matches

A very important distinction is that Hibernica is permissionless in market creation and participation, but the first matching layer may be centralized or semi-centralized.

Hibernica can be permissionless in the following dimensions:

- anyone can create a market,
- anyone can deposit inventory,

- anyone can list an asset market,
- anyone can submit a bid,
- any tokenized asset can theoretically become optionable,
- market creation is not gated by a centralized listing committee.

However, version one may not be fully permissionless in the following dimensions:

- trade matching,
- bid acceptance,
- pricing approval,
- final execution routing,
- solver operation.

The precise claim should therefore be:

Hibernica enables permissionless market origination, with solver-mediated execution.

This is more accurate than claiming that every part of the system is fully decentralized from day one.

14.12. Step Ten: Later Versions Can Automate Execution

In the MVP, the solver operators may manually review bids or run a simple off-chain process.

Later, the process can become more systematic, decentralized and provable.

The automation can use:

- depositor target APR,
- current spot price,
- historical volatility,
- implied volatility estimates,
- time to expiry,
- vault utilization,
- bid ranking,

- price bands,
- stale quote checks,
- oracle validation,
- and market-maker constraints.

A later solver may run something conceptually similar to:

For each vault:

access or decrypt target APR

fetch all valid bids

normalize each bid into APR / implied volatility / edge terms

filter invalid or stale bids

rank bids by depositor outcome

execute highest valid bids until capacity is used

This could run every 24 hours, continuously, or per auction epoch.

15. Dynamic Strikes and Intent-Based Options

Another important concept is that the depositor does not necessarily need to choose fixed strikes upfront.

Instead, they may define:

- target APR,
- duration,
- percentage out-of-the-money,
- or acceptable moneyness band.

Then the system can dynamically form options based on current spot price.

For example:

“Sell 30-day calls approximately 25 percent out of the money, but only if the premium implies at least 40 percent APR.”

If spot moves, the strike band moves with it.

This means the market is not necessarily a static order book of predefined strikes. It can be more intent-based:

Depositor intent:

Sell upside on this asset if yield > X
and strike is sufficiently out-of-the-money.

Market-maker intent:

Buy upside if price, volatility, and oracle constraints are satisfied.

Solver:

Find compatible overlap and execute.

This is more flexible than a traditional listed-option grid.

16. Relationship to Perpetuals and Hedging

The broader Hibernica architecture may also include the ability to pair options with perpetual markets.

The pitch to market makers becomes stronger if they can hedge.

A more advanced structure could give the market maker:

- an option position,
- a corresponding perp hedge or short/long position,
- and potentially special funding mechanics or locked funding conditions.

This would allow a market maker to bid on an option package while being more confident that hedge conditions remain stable.

However, this should be treated as a second-stage product. The simpler Hibernica product is:

Deposit inventory, receive option bids, execute covered-call-style trades.

The hedged option plus perp package is more advanced and should be introduced after users understand the base product.

17. Relationship Between Hibernica and Helix at the Technical Level

In Hibernica:

- Hibernica operates the public marketplace.
- Depositors come to Hibernica.

- Market makers bid into Hibernica.
- Hibernica’s solver coordinates execution.

In Helix:

- a market maker gets its own branded or private version of this stack,
- can quote many markets,
- can integrate its own pricing systems,
- can manage its own clients or vaults,
- can register custom verifiers or oracles,
- and can use the same underlying Hibernica execution infrastructure.

Hibernica is the marketplace.

Helix is the professional quoting and market-maker infrastructure layer.

18. Technical Flow Summary

A clean first-version technical flow could look like this:

1. **Market creation.** A user creates an options vault for a token.
2. **Inventory deposit.** Depositors deposit the token into the vault.
3. **Depositor constraints.** Depositors define target APR, duration, and risk preferences. Sensitive values may be encrypted or kept private.
4. **Vault discovery.** Market makers browse available vaults.
5. **Bid submission.** Market makers submit signed option-buying intents. Each bid includes premium, size, expiry, strike logic, and validity constraints.
6. **Bid validation.** The solver checks signatures, balances, oracle conditions, price bands, and expiry.
7. **Best execution.** The solver compares valid bids against depositor constraints and selects the best bids.
8. **Execution.** The accepted bid creates an option position. Premium is transferred from bidder to vault. Vault inventory is reserved as collateral for the option.
9. **Settlement.** At expiry or exercise, the option is physically or cash settled.
10. **Distribution.** Premium, yield, and any remaining collateral are distributed to vault depositors.

19. Hibernica in the Landscape of Best Execution

The term *BestEx*, or *Best Execution*, traditionally refers to the obligation of a broker, venue, or execution agent to obtain the most favorable reasonably available outcome for a client order under prevailing market conditions. In traditional finance, this concept is operationalized through order-routing policies, venue selection, execution algorithms, transaction cost analysis, post-trade reporting, and periodic execution-quality reviews.

Hibernica approaches the same problem from a different market structure. It is not merely trying to route an order to the best venue. It is creating a new execution environment in which the relevant liquidity, reservation prices, execution constraints, and option bids may not exist in a public order book at all.

In that sense, Hibernica extends the Best Execution concept from *venue selection* to *market formation*.

19.1. Traditional Best Execution

In conventional capital markets, *Best Execution* typically assumes that a client order already exists and that multiple execution venues, liquidity pools, dealers, or market makers may be available. The execution agent must then determine how to route, slice, time, and execute the order in a way that optimizes the client outcome.

Typical *Best Execution* factors include:

- execution price,
- transaction cost,
- speed of execution,
- likelihood of execution,
- likelihood of settlement,
- market impact,
- order size,
- venue quality,
- available liquidity,
- and consistency with the client's instructions.

This model is most natural in mature markets such as equities, futures, FX, and listed options, where venues already exist and liquidity is observable or inferable.

Execution technology providers in this landscape typically focus on:

- smart order routing,
- algorithmic execution,
- execution management systems,
- transaction cost analysis,
- pre-trade cost estimation,
- real-time execution monitoring,
- and post-trade measurement.

The core question in traditional Best Execution is therefore:

Given an order and a set of available execution venues, how do we achieve the best possible execution outcome?

19.2. Why Hibernica Is Structurally Different

Hibernica operates in a different environment.

For many long-tail crypto assets, tokenized assets, real-world assets, and community-held inventories, there may be no listed options market, no continuous order book, no standardized strike surface, and no obvious venue to route to. The market often does not exist before Hibernica creates it.

Therefore, Hibernica’s BestEx solution is not simply routing. It is a combined effort of:

1. market origination,
2. private preference handling,
3. quote solicitation,
4. conditional intent validation,
5. auction-style bid comparison,
6. solver-mediated execution,
7. settlement enforcement,
8. and post-trade auditability.

In traditional Best Execution, the broker asks:

Where should this order be executed?

In Hibernica, the solver asks:

Which private bids, under which constraints, produce the best executable outcome for this vault’s hidden economic preferences?

This makes Hibernica closer to a *Best Execution market-construction layer* than a traditional smart order router.

19.3. Best Execution for Vault-Based Options

In Hibernica, the client-like party is usually the inventory provider: a whale, foundation, treasury, community member, or tokenized asset issuer that deposits tokens into a vault.

The depositor is not submitting a conventional market order. Instead, the depositor is expressing an economic intent:

I am willing to sell optionality on this inventory if the premium produces at least my required yield, subject to my risk preferences.

This intent may be represented through a target APR, duration preference, acceptable money-ness range, utilization limit, or other vault-level constraints.

The market-maker side submits signed conditional bids:

I am willing to buy this option exposure at this premium, provided my execution constraints are satisfied.

The Hibernica solver then performs the Best Execution function by comparing available bids against the depositor’s private constraints and executing the best valid bid or combination of bids.

The goal is not merely to execute above the depositor’s minimum threshold. The goal is to maximize the depositor’s realized outcome among all valid executable bids.

19.4. The Hibernica BestEx Solver

The Hibernica BestEx solver can be understood as the execution agent of the OTC vault system.

Its responsibilities include:

- reading or decrypting the vault’s private economic constraints,
- collecting market-maker bid intents,
- validating bidder signatures,

- checking premium availability,
- checking oracle freshness,
- evaluating bid-specific execution constraints,
- normalizing bids into comparable economic terms,
- ranking valid bids by depositor outcome,
- selecting the best executable bid or bid set,
- executing the option transaction,
- reserving or locking vault collateral,
- transferring premium,
- and recording an auditable execution trace.

Conceptually, the solver executes the following logic:

```

For each vault:
  Read vault inventory and depositor constraints
  Access or decrypt the depositor's private target APR
  Collect all signed market-maker bids
  Filter invalid, expired, stale, or undercollateralized bids
  Check each bidder's execution constraints
  Convert each valid bid into depositor-equivalent yield
  Rank bids by realized depositor outcome
  Execute the highest-value valid bid or bid set
  Record execution evidence for audit and verification

```

This is the core of Hibernica's Best Execution claim.

19.5. Best Execution as Surplus Preservation

The key economic distinction is surplus preservation.

Suppose a depositor is willing to sell a covered call if the implied yield is at least 30% APR. A market maker submits a bid equivalent to 50% APR. A conflicted intermediary could theoretically fill the depositor at 30%, internalize the difference, and retain the 20% spread.

Hibernica's intended BestEx design rejects this model.

Instead, if the best valid bid is 50% APR, the depositor should receive the benefit of that 50% APR execution, minus transparent protocol, solver, or market-creator fees.

In this structure, the solver does not exist to extract hidden spread. It exists to enforce a rule:

Execute at the best available valid market-maker bid, not merely at the depositor's minimum acceptable level.

This is the central Best Execution principle in the Hibernica context.

19.6. Comparison With Traditional BestEx Platforms

Traditional BestEx and algorithmic execution platforms usually optimize execution after a market already exists. They are primarily concerned with routing, scheduling, slicing, market impact, and transaction cost measurement.

Hibernica differs in several dimensions:

Dimension	Traditional BestEx	Hibernica BestEx
Primary market type	Listed or mature markets	Long-tail, OTC, and newly originated options markets
Core problem	Routing an order to the best venue or liquidity source	Creating and executing a market around deposited inventory
Client instruction	Buy or sell a financial instrument	Monetize inventory subject to yield and risk constraints
Liquidity source	Exchanges, dark pools, dealers, market makers	Signed conditional bids from market makers and sophisticated counterparties
Execution object	Order	Intent, vault constraint, and option bid
Optimization target	Price, cost, speed, impact, fill probability	Depositor yield, valid premium, execution quality, collateral safety, and constraint satisfaction
Data visibility	Often venue-level and market-data driven	Mix of public vault data, private depositor thresholds, private bids, and oracle states
Post-trade analysis	Transaction cost analysis and execution-quality review	Execution proof, bid comparison, constraint satisfaction, and surplus attribution
Trust model	Broker or execution provider governance	Solver, smart contracts, cryptographic attestations, and potentially decentralized sequencing

The result is that Hibernica is not competing directly with conventional BestEx systems as an execution algorithm for listed markets. It is applying the Best Execution principle to a new class of markets where the option venue, liquidity surface, and execution process are created by the protocol itself.

19.7. Role of TEEs in Hibernica Best Execution

A trusted execution environment, or TEE, can materially strengthen the Hibernica BestEx architecture.

The central challenge is that Hibernica may need to operate on private information:

- the depositor’s hidden target APR,
- market-maker bid details,
- bidder-specific constraints,
- model-derived validity checks,
- and solver ranking logic.

Without additional guarantees, users must trust the solver operator not to leak information, reorder bids unfairly, selectively ignore better bids, or execute in a way that favors one counterparty.

A TEE can reduce this trust assumption by allowing the matching logic to run inside an attested environment. The TEE can receive encrypted depositor thresholds and private bid data, run the prescribed BestEx algorithm, and produce an attestation that the correct code was executed.

A TEE-based BestEx system could provide the following guarantees:

- depositor thresholds remain hidden from market makers,
- market-maker bids remain hidden from competing bidders until execution or expiry,
- the solver cannot arbitrarily change the ranking algorithm without changing the attested code hash,
- the winning bid can be proven to satisfy the vault’s constraints,
- rejected bids can be proven invalid or inferior under the ranking rule,
- and the execution trace can be made auditable without exposing all private inputs.

This allows Hibernica to move from a trust-based Best Execution claim to a verifiable Best Execution claim.

The architectural statement is:

The TEE converts Best Execution from an operational promise into an attested computation over private market data.

19.8. Role of a Zellular aBFT Sequencer

A Zellular-style asynchronous Byzantine fault tolerant sequencer can further improve the execution architecture by addressing ordering, availability, censorship resistance, and multi-solver coordination.

In a solver-mediated OTC system, ordering matters. If bids arrive over time and the solver can choose which bids to see, include, exclude, or sequence, then execution fairness depends on the integrity of the ordering layer.

An aBFT sequencer can provide a more neutral ordering and availability layer for intents.

Its role would be to:

- receive encrypted or committed market-maker bids,
- timestamp and order bid submissions,
- prevent unilateral censorship by one solver operator,
- make bid inclusion auditable,
- provide a canonical input set for each auction or execution epoch,
- coordinate multiple solver or TEE instances,
- and reduce the ability of any single operator to manipulate execution sequencing.

In this model, the aBFT sequencer does not need to know the private contents of each bid. It can order encrypted bid commitments and provide a canonical data-availability layer. The TEE or solver then decrypts or accesses the relevant data and runs the BestEx matching algorithm over the ordered input set.

The combined architecture is therefore:

aBFT sequencer for fair ordering and availability.
TEE for private computation and attested BestEx logic.
Smart contracts for settlement, collateral, and final state enforcement.

19.9. TEE Plus aBFT BestEx Architecture

The strongest version of Hibernica’s Best Execution architecture combines four layers:

1. **Vault layer.** Depositors provide inventory and encrypted or committed economic constraints.
2. **Intent layer.** Market makers submit signed conditional bids.

3. **Sequencing layer.** A Zellular-style aBFT sequencer orders bids, provides availability, and defines the canonical input set.
4. **Private execution layer.** A TEE runs the BestEx algorithm over private vault constraints and bid data.
5. **Settlement layer.** Smart contracts enforce premium transfer, collateral reservation, option creation, expiry, and settlement.

The flow can be expressed as:

1. Depositor creates vault and commits encrypted target APR.
2. Market makers submit signed encrypted bids.
3. aBFT sequencer orders bid commitments for the auction epoch.
4. TEE receives the canonical input set.
5. TEE decrypts or accesses private bid and vault data.
6. TEE validates signatures, balances, oracles, and constraints.
7. TEE ranks all valid bids according to the public BestEx rule.
8. TEE outputs the winning execution instruction and attestation.
9. Smart contracts execute premium transfer and option creation.
10. Execution evidence is recorded for later audit.

This creates a credible path toward verifiable OTC best execution without revealing sensitive trading information.

19.10. Why This Is Superior to a Purely Centralized Solver

A purely centralized solver can deliver good execution, but users must trust the operator.

The operator could theoretically:

- see depositor thresholds,
- leak bid information,
- prefer affiliated market makers,
- ignore superior bids,
- reorder submissions,
- execute stale bids opportunistically,
- or extract hidden spread.

A TEE and aBFT sequencing stack reduces these risks.

The TEE constrains what the solver can do with private data.

The aBFT sequencer constrains which bids the solver must consider and in what canonical order.

The settlement contracts constrain what can be executed on-chain.

Together, these components create a layered trust model:

The solver computes.
The TEE attests.
The sequencer orders.
The contracts enforce.

This is a stronger claim than simply saying that Hibernica provides best execution. It allows Hibernica to say that it provides *verifiable, privacy-preserving best execution for OTC options markets*.

19.11. Best Execution Without Public Price Discovery Leakage

A major difficulty in OTC options is that best execution and privacy are often in tension.

If all bids are public, then price discovery is transparent, but market makers may be reluctant to expose their edge, models, or axes.

If all bids are private, then market makers are protected, but depositors must trust the intermediary to choose the best bid.

Hibernica’s architecture aims to resolve this tension.

With encrypted bids, TEE-based computation, and a sequenced input set, the system can preserve privacy while still proving that the winning bid was selected according to the declared BestEx rule.

This is especially important for long-tail options because public order books may be too thin, too easy to manipulate, or too revealing for sophisticated counterparties.

Hibernica’s BestEx model is therefore not a public central-limit-order-book model. It is closer to a cryptographically assisted RFQ auction model:

Private bids in, attested best valid execution out.

19.12. Potential BestEx Metrics for Hibernica

To make the Best Execution claim measurable, Hibernica should define explicit execution-quality metrics.

Possible metrics include:

- **Premium improvement:** realized premium versus depositor minimum.
- **APR improvement:** realized APR versus target APR.
- **Bid coverage:** number of valid bids considered per vault or epoch.
- **Execution surplus:** economic surplus returned to the depositor rather than captured by the intermediary.
- **Constraint satisfaction:** proof that executed trades satisfied depositor and bidder constraints.
- **Bid rejection reason:** invalid signature, stale oracle, insufficient premium, expired bid, inferior price, or constraint failure.
- **Time to execution:** elapsed time between auction close and option creation.
- **Fill quality:** executed notional relative to available vault inventory and valid bid demand.
- **Settlement quality:** successful expiry, exercise, collateral release, and payoff settlement.
- **Attestation coverage:** percentage of executed trades processed through attested BestEx logic.

Over time, these metrics could form Hibernica’s equivalent of transaction cost analysis.

However, unlike traditional TCA, which often compares an execution to market benchmarks such as arrival price, VWAP, TWAP, or implementation shortfall, Hibernica’s execution analysis would compare outcomes against vault-specific private constraints and competing signed bids.

19.13. Strategic Positioning

Hibernica should avoid presenting BestEx merely as a branding term. The stronger framing is that Hibernica introduces a new category:

Best Execution for permissionless OTC derivatives origination.

The traditional BestEx industry optimizes execution in markets that already exist.

Hibernica creates the market, solicits the bids, validates private constraints, executes the best valid option trade, and settles the resulting position.

This gives Hibernica a differentiated claim:

Traditional BestEx routes orders across existing liquidity. Hibernica BestEx creates liquidity around idle inventory and proves that the best valid bid was executed.

With TEE and Zellular aBFT infrastructure, that claim can become even stronger:

Hibernica can deliver privacy-preserving, censorship-resistant, auditable Best Execution for OTC options on any tokenized asset.

20. The Simplest Mental Model

Anyone can deposit tokens.
Anyone can bid to buy options on those tokens.
The solver accepts the best valid bids.
Depositors earn premium.
Market makers get long-tail volatility exposure.

That is the core product.